

V4: Architecture, Training Objectives, and Retrieval

Detailed methods draft

Implementation snapshot: 21 September 2026

Abstract

We describe a bidirectional enzyme–reaction retrieval method with independently computable protein and reaction representations. The protein encoder combines a global sequence representation, a frozen SLEEC residue prior, and four learned residue views. The reaction encoder combines frozen reaction, molecular, and chemical features. Training comprises minibatch, anchor-balanced multi-positive alignment followed by full-graph residual refinement. The final retrieval score combines the refined neural representation with a fixed dictionary of training reactions. An optional biological objective regularizes the existing refinement heads using EC hierarchy, cofactor-associated categories, and coarse transformation descriptors, without introducing additional inference parameters. This document specifies the original V4 recipe and its phase-2-only biological variant, including the exact distinction between their training and inference computations.

Contents

1	Method	3
1.1	Problem formulation and overview	3
1.2	Protein encoder: global, SLEEC, and learned residue views	3
1.3	Reaction encoder: molecular sides and modality fusion	5
1.4	Phase 1: anchor-balanced, decoupled multi-positive alignment	6
1.5	Phase 2: full-graph residual refinement	7
1.6	Biological supervision without additional encoder components	8
1.7	Training-reaction dictionary representation	10
1.8	Inference adjustments and final bidirectional score	10
1.9	Optimization and target-specific fitting	11
A	Notation and exact layer specifications	12
B	Attention implementation and sequence preprocessing	13
C	Fixed chemistry descriptor	14
D	Efficient evaluation of the biological objective	14
E	Why the two association losses differ	15
F	Annotation provenance and benchmark-specific inputs	15
G	Optimization recipe and executable order	16

1 Method

1.1 Problem formulation and overview

Let $\mathcal{R}_{\text{tr}} = \{r_1, \dots, r_{N_R}\}$ and $\mathcal{E}_{\text{tr}} = \{e_1, \dots, e_{N_E}\}$ denote the unique reaction and enzyme identifiers appearing in a target dataset’s training split. The observed association graph is $\mathcal{D}_{\text{tr}} \subseteq \mathcal{R}_{\text{tr}} \times \mathcal{E}_{\text{tr}}$, with binary adjacency $Y_{ij} = \mathbb{I}[(r_i, e_j) \in \mathcal{D}_{\text{tr}}]$. A reaction can have several associated enzymes and an enzyme can have several associated reactions. We learn a score $S(r, e)$ used to rank enzymes for a reaction ($R \rightarrow E$) or reactions for an enzyme ($E \rightarrow R$). The score is a retrieval compatibility measure, rather than a calibrated probability of catalytic activity.

V4has independent protein and reaction encoders. Phase 1 learns 512-dimensional neural representations from the target training associations. Phase 2 freezes these encoders and learns a small residual refinement for each endpoint using the complete target training graph. At inference, the refined representations are combined with a fixed representation whose coordinates index training reactions (Figure 1). Candidate vectors can therefore be precomputed: encoding a protein does not require the current reaction query, and encoding a reaction does not require the current screening library.

We describe two objectives for the same architecture. Original V4uses association supervision and geometry preservation in phase 2. V4+Bioadds a relative biological neighborhood loss to those same heads. For the reported `all_1` configuration, biological supervision is used only in phase 2. The protein residue queries, SLEEC scorer, reaction encoder, and dictionary remain fixed during this addition.

Throughout, $n(x) = x / \max(\|x\|_2, \epsilon)$ denotes numerical ℓ_2 normalization, $[x]_+ = \max(0, x)$, $[x; y]$ denotes concatenation, and LN denotes layer normalization. All attention distributions exclude padding. Distinct occurrences of a learnable projection have independent parameters unless sharing is explicitly stated.

1.2 Protein encoder: global, SLEEC, and learned residue views

For an enzyme sequence e with L retained residues, a frozen ProtT5 model provides contextual residue vectors $H_e = [h_1, \dots, h_L]^\top \in \mathbb{R}^{L \times 1024}$ [2]. We construct six protein views: one global view, one view defined by a frozen SLEEC functional-residue scorer [3], and $K = 4$ learned residue views. The scorer is a pretrained resource; it is not fitted to each retrieval target in phase 1.

Global and SLEEC views. The global view is the mean of the retained residue embeddings,

$$x_e^{(0)} = \frac{1}{L} \sum_{i=1}^L h_i. \quad (1)$$

The frozen scorer computes a scalar logit ℓ_i for each residue using a $1024 \rightarrow 256 \rightarrow 1$ MLP with a ReLU hidden activation. We center these logits within the protein, clip them, and form a smoothed attention distribution:

$$\bar{\ell} = L^{-1} \sum_{i=1}^L \ell_i, \quad t_i = \text{clip}(\ell_i - \bar{\ell}, -10, 10), \quad (2)$$

$$p_i^S = \frac{\exp(t_i)}{\sum_{j=1}^L \exp(t_j)}, \quad a_i^S = (1 - \eta)p_i^S + \frac{\eta}{L}, \quad \eta = 0.05, \quad (3)$$

$$x_e^{(1)} = \sum_{i=1}^L a_i^S h_i. \quad (4)$$

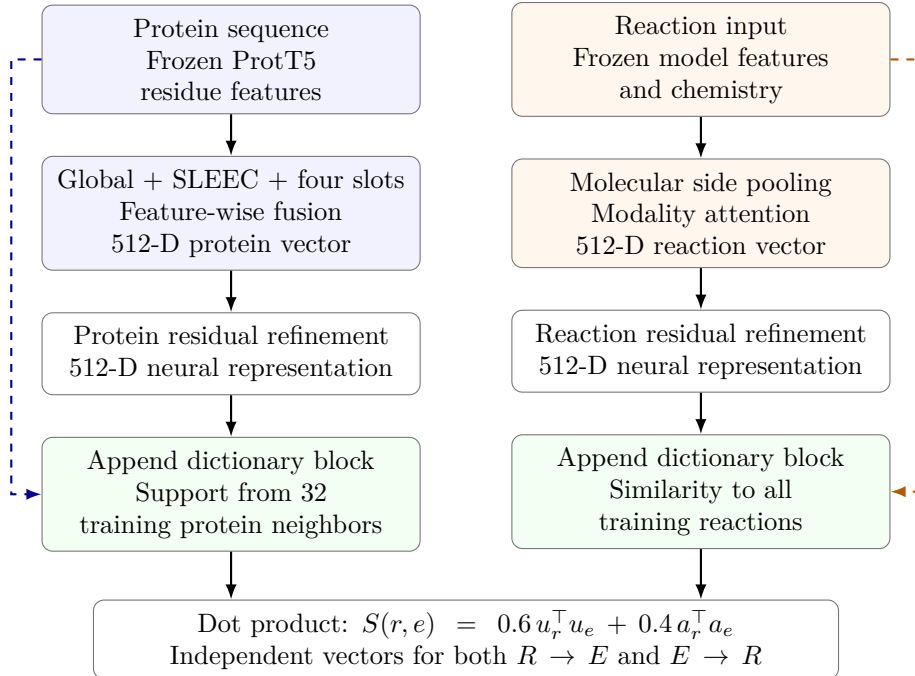


Figure 1: Inference structure of V4. Dashed paths supply frozen raw features to the dictionary maps. Dictionary coordinates are computed from frozen raw features and training associations, independently of the neural refinement heads. The arrows into the appended representations indicate composition, not that dictionary features are generated from the heads. The optional biological loss acts on the two residual heads during phase 2; it introduces no additional inference branch. The protein fused-view multiplier and refinement multiplier are specified in Section 1.8.

Thus, SLEEC contributes a soft residue-weighted view. The threshold used for SLEEC diagnostics does not select or remove residues in this encoder.

Four learned residue views. A shared residue adapter maps each contextual residue to $d_h = 256$ dimensions:

$$\tilde{h}_i = \text{GELU}(W_a \text{LN}(h_i) + b_a), \quad k_i = \text{n}(W_k \tilde{h}_i), \quad v_i = W_v \tilde{h}_i + b_v. \quad (5)$$

Here W_k has no bias. The model contains four shared trainable query vectors $q_k \in \mathbb{R}^{256}$, initialized orthogonally. They are model parameters, not reaction-dependent queries. For slot k , attention is

$$\gamma = 10 \text{sigmoid}(\rho_\gamma), \quad \gamma_{\text{init}} = 4, \quad (6)$$

$$p_{ki} = \frac{\exp(\gamma \text{n}(q_k)^\top k_i)}{\sum_{j=1}^L \exp(\gamma \text{n}(q_k)^\top k_j)}, \quad a_{ki} = (1 - \eta)p_{ki} + \frac{\eta}{L}, \quad (7)$$

$$x_e^{(k+1)} = \sum_{i=1}^L a_{ki} v_i, \quad k \in \{1, \dots, 4\}. \quad (8)$$

Each slot can select a different content-dependent summary of the same protein. The four views are fused into one protein vector; they are not four independently scored enzyme prototypes. SLEEC defines its own view and is not added as a common logit bias to the four learned slots.

Feature-wise view fusion. Each view is projected to 256 dimensions by an independent network

$$F_{a \rightarrow b}(x) = W_2 \text{Drop}_{0.1}(\text{GELU}(W_1 \text{LN}(x) + b_1)) + b_2, \quad (9)$$

whose hidden width is 512 and whose training dropout probability is 0.1. For the first two views $a = 1024$; for learned slots $a = 256$. Define $y_e^{(v)} = \text{n}(F_v(x_e^{(v)}))$ for $v = 0, \dots, 5$. A $1536 \rightarrow 256 \rightarrow 1536$ gate MLP with GELU maps their concatenation to logits $G_e \in \mathbb{R}^{6 \times 256}$. The gate softmax is taken over views *separately for each feature coordinate*:

$$\omega_{vj} = (1 - \zeta) \frac{\exp(G_{vj})}{\sum_{u=0}^5 \exp(G_{uj})} + \frac{\zeta}{6}, \quad \zeta = 0.05, \quad (10)$$

$$\bar{y}_{e,j} = \sum_{v=0}^5 \omega_{vj} y_{e,j}^{(v)}. \quad (11)$$

The final gate layer is initialized to zero, giving equal initial view weights. The floor is $\zeta/6$ per view and coordinate, not 0.05 per view.

Two additional networks of the form in Eq. (9) produce separately normalized global and fused outputs in $\mathbb{R}^{512}$:

$$g_e = \text{n}(F_g(x_e^{(0)})), \quad f_e = \text{n}(F_f(\bar{y}_e)), \quad (12)$$

$$s = 0.5 \text{sigmoid}(\rho_s), \quad b_e = \text{n}(g_e + s f_e). \quad (13)$$

The scalar s is shared across proteins, is initialized to 0.1, and remains strictly between 0 and 0.5. The direct global branch provides a stable route from the mean sequence features while the fused branch learns a correction.

Sequence order and interpretation. No additional positional encoding is applied in the pooling module. Its queries attend to the contextual content of ProtT5 vectors. Permuting already computed residue vectors and their masks together leaves this pooling operation unchanged, whereas changing the amino-acid sequence can change the contextual vectors. Neither a slot nor a coordinate of the final 512-dimensional vector is assigned a named catalytic function. Attention weights are learned aggregation weights, not validated active-site labels.

1.3 Reaction encoder: molecular sides and modality fusion

We use four feature sources: a 768-dimensional ReactionT5v2 reaction embedding [5, 6], 768-dimensional Uni-Mol2 molecule embeddings [4], 256-dimensional ChIRo molecule embeddings [1], and a 617-dimensional fixed chemistry vector. Their feature extractors are frozen; molecular pooling, modality projections, and fusion are trainable.

Side-specific molecule pooling. For molecular modality $m \in \{\text{UniMol2}, \text{ChIRo}\}$ and side $s \in \{-, +\}$, let $x_{s,j}^m$ be the embedding of molecule j . Independent linear attention scorers for the two sides produce

$$\pi_{s,j}^m = \frac{\exp((w_s^m)^\top x_{s,j}^m + b_s^m)}{\sum_t \exp((w_s^m)^\top x_{s,t}^m + b_s^m)}, \quad p_s^m = \sum_j \pi_{s,j}^m x_{s,j}^m. \quad (14)$$

The configured `directional_delta` composition is

$$c_r^m = [p_-^m; p_+^m; p_+^m - p_-^m; |p_+^m - p_-^m|]. \quad (15)$$

It has dimension 3072 for Uni-Mol2 and 1024 for ChIRo. Direction is represented by the ordered sides and signed difference when physical sides are supplied.

Input semantics. For EnzymeMap, inputs retain their physical reactant and product sides. ReactZyme’s released participant-set inputs use the established self-reaction adapter, placing the same complete set on both sides. The two side poolers still have different parameters, so their pooled vectors need not be equal. Their difference on these self-reactions must not be interpreted as an observed chemical transformation direction. The choice of chemical input representation is separate from the choice of retrieval direction.

Modality tokens and attention. For $m \in \{\text{T5, UniMol2, ChIRo, chem}\}$, let c_r^m denote the corresponding reaction input. Each independent modality MLP has widths $d_m \rightarrow 1024 \rightarrow 512 \rightarrow 512$, with ReLU after the first two linear layers. A shared layer-normalization module then gives the modality token

$$t_r^m = \text{LN}(\text{MLP}_m(c_r^m)) \in \mathbb{R}^{512}. \quad (16)$$

An attention MLP of widths $512 \rightarrow 512 \rightarrow 1$, with a ReLU hidden activation, scores the tokens. If $\mathcal{M}_r$ is the set of available modalities,

$$\alpha_r^m = \frac{\exp(A(t_r^m))}{\sum_{n \in \mathcal{M}_r} \exp(A(t_r^n))}, \quad t_r = \sum_{m \in \mathcal{M}_r} \alpha_r^m t_r^m, \quad b_r = \mathbf{n}(F_r(t_r)). \quad (17)$$

The output MLP F_r has widths $512 \rightarrow 1024 \rightarrow 512$ and a ReLU hidden activation. Unavailable modalities are masked out of attention. Reaction modality dropout and projection dropout are zero in the reported recipe. The resulting b_r and b_e inhabit a shared 512-dimensional retrieval space.

1.4 Phase 1: anchor-balanced, decoupled multi-positive alignment

Training minibatches sample association rows and deduplicate their endpoint identifiers before constructing a reaction-by-enzyme score matrix. Let $\mathcal{R}_B$ and $\mathcal{E}_B$ be these distinct endpoints. Positives are identified from *all known training associations between endpoints in the batch*, rather than only the association rows sampled for that batch. For a reaction anchor r , define

$$P_B(r) = \{e \in \mathcal{E}_B : (r, e) \in \mathcal{D}_{\text{tr}}\}, \quad N_B(r) = \mathcal{E}_B \setminus P_B(r), \quad (18)$$

and define $P_B(e)$ and $N_B(e)$ analogously for enzyme anchors. Only anchors with at least one positive and one remaining candidate enter the respective directional average.

The logits are $z_{re} = \beta b_r^\top b_e$ with fixed inverse temperature $\beta = 5$ (temperature 0.2). For each known positive p of an anchor a , the decoupled loss is

$$\ell_{\text{dec}}(a, p) = -\log \frac{\exp(z_{ap})}{\exp(z_{ap}) + \sum_{n \in N_B(a)} \exp(z_{an})} \quad (19)$$

$$= \log \left(1 + \sum_{n \in N_B(a)} \exp(z_{an} - z_{ap}) \right). \quad (20)$$

Other known positives are excluded from the denominator for this individual positive. Thus every positive must compete with the remaining candidates, without competing with its known positive partners. This differs from a single log-sum over all positives, which can be dominated by one easy positive.

Let $\mathcal{A}_R$ and $\mathcal{A}_E$ be the valid reaction and enzyme anchors in the batch. We first average over positives within an anchor, then average over anchors, and finally average the two directions:

$$\mathcal{L}_{R \rightarrow E}^{(1)} = \frac{1}{|\mathcal{A}_R|} \sum_{r \in \mathcal{A}_R} \frac{1}{|P_B(r)|} \sum_{e \in P_B(r)} \ell_{\text{dec}}(r, e), \quad (21)$$

$$\mathcal{L}_{E \rightarrow R}^{(1)} = \frac{1}{|\mathcal{A}_E|} \sum_{e \in \mathcal{A}_E} \frac{1}{|P_B(e)|} \sum_{r \in P_B(e)} \ell_{\text{dec}}(e, r), \quad (22)$$

$$\mathcal{L}_{\text{align}}^{(1)} = \frac{1}{2} \mathcal{L}_{R \rightarrow E}^{(1)} + \frac{1}{2} \mathcal{L}_{E \rightarrow R}^{(1)}. \quad (23)$$

A direction with no valid anchor contributes zero. The normalization prevents an anchor’s contribution from growing simply in proportion to its number of positives in the matrix. It does not make anchors equally frequent over an epoch when association rows are sampled. Unobserved pairs are still used as contrastive negatives with weight 1; they are not measured inactive pairs.

Attention regularization. The protein encoder also returns two small penalties on the *raw* learned-slot distributions p_{ki} , before the uniform mixture in Eq. (8). For a protein of length L , define

$$H_k = - \sum_i p_{ki} \log p_{ki}, \quad H_{\min}(L) = \min\{\frac{1}{2} \log L, \log 4\}, \quad (24)$$

$$E(e) = \frac{1}{K} \sum_{k=1}^K [H_{\min}(L) - H_k]_+^2. \quad (25)$$

This penalty discourages attention with very small effective support. To discourage nearly identical attention patterns, center each raw distribution against uniform attention: $\bar{p}_k = p_k - L^{-1} \mathbf{1}$. For eligible unordered pairs $\mathcal{Q}_e = \{(k, l) : k < l, \|\bar{p}_k\|_2 > 10^{-6}, \|\bar{p}_l\|_2 > 10^{-6}, L > 1\}$, let

$$D(e) = \frac{1}{\max(1, |\mathcal{Q}_e|)} \sum_{(k, l) \in \mathcal{Q}_e} \left[\frac{\bar{p}_k^\top \bar{p}_l}{\|\bar{p}_k\|_2 \|\bar{p}_l\|_2} - 0.9 \right]_+^2. \quad (26)$$

Uniform heads have no meaningful centered direction and are excluded from this comparison. The complete phase-1 objective is

$$\boxed{\mathcal{L}^{(1)} = \mathcal{L}_{\text{align}}^{(1)} + 0.01 \frac{1}{|\mathcal{E}_B|} \sum_{e \in \mathcal{E}_B} E(e) + 0.01 \frac{1}{|\mathcal{E}_B|} \sum_{e \in \mathcal{E}_B} D(e).} \quad (27)$$

There is no EC/cofactor/mechanism loss in this phase for the phase-2-only V4+Bioconfiguration. The safeguards in Eqs. (25) and (26) encourage numerical and representational diversity; they do not establish a biological interpretation of the attention heads.

1.5 Phase 2: full-graph residual refinement

After selecting the epoch-10 phase-1 snapshot, we freeze both F3 encoders and cache their unadjusted outputs b_r, b_e for the unique training endpoints. Two independent residual heads h_R, h_E each comprise layer normalization, a $512 \rightarrow 1024$ linear layer, GELU, and a $1024 \rightarrow 512$ linear layer. The last layer’s weights and biases are initialized to zero. During phase-2 training,

$$u_r^{\text{tr}} = \text{n}(b_r + \kappa h_R(b_r)), \quad u_e^{\text{tr}} = \text{n}(b_e + \kappa h_E(b_e)), \quad \kappa = 0.2. \quad (28)$$

Consequently, both heads initially preserve the normalized base representations. No gradient is propagated to the frozen F3 encoders.

Uniform-positive cross-entropy over the complete graph. We form the full training matrix $Z_{ij} = (u_{r_i}^{\text{tr}})^\top u_{e_j}^{\text{tr}} / \tau_2$, with $\tau_2 = 0.2$. Let $d_i^R = \sum_j Y_{ij}$ and $d_j^E = \sum_i Y_{ij}$. Every retained training endpoint has at least one association. The directional objectives are

$$\mathcal{L}_{R \rightarrow E}^{(2)} = \frac{1}{N_R} \sum_i \left[\log \sum_{j=1}^{N_E} \exp(Z_{ij}) - \frac{1}{d_i^R} \sum_j Y_{ij} Z_{ij} \right], \quad (29)$$

$$\mathcal{L}_{E \rightarrow R}^{(2)} = \frac{1}{N_E} \sum_j \left[\log \sum_{i=1}^{N_R} \exp(Z_{ij}) - \frac{1}{d_j^E} \sum_i Y_{ij} Z_{ij} \right], \quad (30)$$

$$\mathcal{L}_{\text{CE}}^{(2)} = \frac{1}{2} \mathcal{L}_{R \rightarrow E}^{(2)} + \frac{1}{2} \mathcal{L}_{E \rightarrow R}^{(2)}. \quad (31)$$

These are cross-entropies with a uniform target distribution over each anchor’s known positives. Unlike phase 1, the denominator includes all opposite-type training endpoints, including all positives. Phase 2 therefore changes both the coverage of the contrastive bank and the form of the loss; it is not simply Eq. (20) evaluated on a larger batch.

Preserving the original geometry. An identity penalty limits changes to the frozen base representations:

$$\mathcal{L}_{\text{id}} = \frac{1}{2N_R} \sum_i \left[1 - (u_{r_i}^{\text{tr}})^\top \mathbf{n}(b_{r_i}) \right] + \frac{1}{2N_E} \sum_j \left[1 - (u_{e_j}^{\text{tr}})^\top \mathbf{n}(b_{e_j}) \right]. \quad (32)$$

The original V4phase-2 objective is $\mathcal{L}_{\text{CE}}^{(2)} + 10\mathcal{L}_{\text{id}}$. Both heads are optimized jointly for 100 updates, each of which includes the entire compact training graph.

1.6 Biological supervision without additional encoder components

The biological extension modifies the training objective of these same heads. It constrains relationships among reaction representations and, separately, among enzyme representations. The association loss connects the two endpoint types. No classification heads, biological feature coordinates, additional residue queries, or inference-time annotation inputs are introduced.

Annotation families and confidence. We use three families $\mathcal{F} = \{\text{EC}, \text{cof}, \text{mech}\}$. Each training endpoint can belong to multiple categories. A membership in category c has confidence $w_{ic} \in (0, 1]$; missing memberships create no annotation evidence. For EC, each specified prefix is retained. A category at hierarchy depth $d \in \{1, 2, 3, 4\}$ has weight $\xi_c \in \{0.125, 0.25, 0.5, 1\}$, respectively. Unknown EC digits are not imputed. Cofactor-associated categories are recognized from reaction participants and assigned confidence 0.4. They describe participant presence, which is an imperfect proxy for a biochemical cofactor requirement. Mechanism categories are coarse descriptors derived from atom-mapped bond changes; mappings below confidence 0.5 are omitted. These descriptors do not specify complete catalytic mechanisms. Cofactor and mechanism category weights are $\xi_c = 1$.

Reaction annotations are transferred to proteins only along training associations. When several observations support the same membership, the maximum confidence is retained. Native protein EC annotations can additionally describe multiple functions of a training protein. Appendix F specifies the different annotation sources for the two benchmarks.

Relative within-category and background distances. Consider one endpoint type $X \in \{R, E\}$ and one family f . Let $\hat{z}_i = \mathbf{n}(u_i^{\text{tr}})$ and $\delta_{ij} = 1 - \hat{z}_i^\top \hat{z}_j$. Let I_c be the endpoints annotated with category c , $n_c = |I_c|$, and $\mathcal{A}_f = \bigcup_{c \in \mathcal{F}} I_c$. The comparison set is $B_c = \mathcal{A}_f \setminus I_c$. An endpoint without any annotation in f is excluded from both membership and background for that family. For a multiply annotated background endpoint, set $\bar{w}_j^f = \max_{c' \in f: j \in I_{c'}} w_{jc'}$.

For eligible categories, define the within-category and background confidence masses

$$T_c = \sum_{\substack{i, j \in I_c \\ i \neq j}} w_{ic} w_{jc}, \quad U_c = \sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f, \quad (33)$$

and confidence-weighted mean distances

$$d_c^{\text{within}} = \frac{1}{T_c} \sum_{\substack{i, j \in I_c \\ i \neq j}} w_{ic} w_{jc} \delta_{ij}, \quad (34)$$

$$d_c^{\text{background}} = \frac{1}{U_c} \sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f \delta_{ij}. \quad (35)$$

The per-category penalty is

$$\ell_c = \underbrace{\frac{T_c}{n_c(n_c - 1)}}_{\text{mean pair confidence}} \left[m + d_c^{\text{within}} - d_c^{\text{background}} \right]_+, \quad m = 0.1. \quad (36)$$

Its prefactor preserves the strength of annotation confidence. For example, if every membership in a category has confidence 0.4, this prefactor is $0.4^2 = 0.16$. Computing only confidence-normalized distances would remove that common confidence reduction.

A category contributes only if $n_c \geq 2$, $|B_c| \geq 1$, $T_c > 0$, and $U_c > 0$. Let $\mathcal{C}_{X,f}^*$ be the eligible categories. We use a category-weighted average, followed by a fixed average over endpoint types:

$$\mathcal{L}_{X,f} = \frac{\sum_{c \in \mathcal{C}_{X,f}^*} \xi_c \ell_c}{\sum_{c \in \mathcal{C}_{X,f}^*} \xi_c}, \quad \mathcal{L}_f = \frac{1}{2} (\mathcal{L}_{R,f} + \mathcal{L}_{E,f}). \quad (37)$$

An empty eligible set contributes zero. Averaging by category, rather than by the total number of pairs across categories, limits domination by very large annotation groups. Membership overlap is allowed; the background for one category can share other categories with its members.

Complete biological phase-2 objective. For V4+Bio, the exact objective is

$$\mathcal{L}^{(2)} = \mathcal{L}_{\text{CE}}^{(2)} + 10\mathcal{L}_{\text{id}} + \frac{1}{3} \sum_{f \in \mathcal{F}} \lambda_f \mathcal{L}_f. \quad (38)$$

The `all_1` configuration sets all three $\lambda_f = 1$; original V4sets all three to zero. In family-removal controls, only the corresponding coefficient is set to zero; the divisor remains three. Shuffled-label controls preserve annotation profiles, confidences, and category frequencies while permuting their assignment to training endpoints.

The relative loss expresses a geometric preference for annotated peers. The background does not assert that a protein is inactive for a reaction, and different categories are not mutually exclusive biological functions. Incomplete annotations and enzyme promiscuity can make these geometric references imperfect. The relative hinge does not by itself guarantee non-collapsed embeddings; association supervision remains necessary. In the phase-2-only variant, biological gradients reach the residual heads but not the frozen residue queries or feature extractors. Exact category-sum identities avoid constructing a dense biological pairwise matrix (Appendix D).

1.7 Training-reaction dictionary representation

In addition to the neural embedding, both endpoints receive a vector $a \in \mathbb{R}^{N_R}$ whose coordinates index the target’s training reactions. The dictionary uses frozen raw features and observed training associations. It is fitted once for each target split and is independent of the current query or candidate collection.

Training-only centering. For a raw feature family with available training values x_i , we compute

$$\mu = \frac{1}{N_{\text{avail}}} \sum_{i \in \text{available train}} \mathbf{n}(x_i), \quad \psi(x) = \mathbf{n}(\mathbf{n}(x) - \mu). \quad (39)$$

The order is normalization, centering, then normalization. Enzyme features are cached ProtT5 residue means. Reaction features concatenate four separately centered and normalized blocks (ReactionT5v2, Uni-Mol2, ChIRo, chemistry), followed by normalization of the concatenation. Unavailable blocks are zero. For the dictionary’s molecular blocks, the implemented export uses the *mean forward-reactant bank*: this is the complete participant set for ReactZyme self-reactions and the reactant side for EnzymeMap. These fixed raw features differ from the trainable side pooling in the neural reaction tower.

Enzyme-to-training-reaction transfer. Let $\psi_E(e)$ be the centered enzyme feature and let $\mathcal{N}_{32}(e)$ be its 32 nearest training proteins by cosine similarity $c_{ej} = \psi_E(e)^\top \psi_E(e_j)$. With $\tau_E = 0.03$, define

$$w_{ej}^{\text{NN}} = \exp\left(\frac{c_{ej} - \max_{t \in \mathcal{N}_{32}(e)} c_{et}}{\tau_E}\right), \quad (40)$$

$$\tilde{a}_{e,i} = \max_{\substack{j \in \mathcal{N}_{32}(e) \\ Y_{ij}=1}} w_{ej}^{\text{NN}}, \quad a_e = \mathbf{n}(\tilde{a}_e). \quad (41)$$

An empty maximum is zero. The maximum merges repeated support for a training reaction without summing the frequency of similar supporting proteins. There is no exclusion of an exact training protein when it is an eligible neighbor. Such information is permitted by the benchmark’s training graph.

Reaction-to-training-reaction response. Let $\psi_R(r)$ denote the normalized raw reaction feature and $c_{ri} = \psi_R(r)^\top \psi_R(r_i)$. We use all training reaction anchors:

$$\tilde{a}_{r,i} = \exp\left(\frac{c_{ri} - \max_t c_{rt}}{\tau_R}\right), \quad a_r = \mathbf{n}(\tilde{a}_r), \quad \tau_R = 0.03. \quad (42)$$

The implementation floors full-support exponential responses at the smallest positive normal FP32 value before normalization. The rowwise maximum in Eqs. (41)–(42) stabilizes exponentiation and otherwise cancels under normalization. Full reaction-anchor support avoids exact support mismatch caused by truncating both endpoint vectors to separate nearest-anchor sets. It does not ensure informative transfer for a reaction remote from the training chemistry.

1.8 Inference adjustments and final bidirectional score

The reported recipe makes two explicit inference adjustments after both training stages. First, it strengthens the entire fused protein branch:

$$\tilde{b}_e = \mathbf{n}(g_e + \nu s f_e), \quad \nu = 3, \quad \tilde{b}_r = b_r. \quad (43)$$

The scalar νs is checked to remain below 0.5 for these checkpoints. This multiplies the complete fused-view contribution, including its global, SLEEC, and learned-slot information. It does not multiply only SLEEC weights. Phase 2 was fitted to the native b_e , before this adjustment.

Second, the existing residual heads are applied with inference multiplier $c = 0.5$:

$$u_e = n(\tilde{b}_e + c\kappa h_E(\tilde{b}_e)), \quad u_r = n(b_r + c\kappa h_R(b_r)), \quad c\kappa = 0.1. \quad (44)$$

The multiplier halves the residual coefficient relative to training. It is not a hard bound on the update norm or on the resulting angular displacement. The dictionary representation is unchanged by either adjustment.

With semantic weight $\alpha = 0.4$, define independent endpoint embeddings

$$\Phi_E(e) = [\sqrt{1-\alpha} u_e; \sqrt{\alpha} a_e], \quad \Phi_R(r) = [\sqrt{1-\alpha} u_r; \sqrt{\alpha} a_r]. \quad (45)$$

Their dot product gives the final score

$$S(r, e) = \Phi_R(r)^\top \Phi_E(e) = 0.6 u_r^\top u_e + 0.4 a_r^\top a_e. \quad (46)$$

Thus the neural block has 512 dimensions, whereas the complete representation has $512 + N_R$ dimensions. The same score is used in both retrieval directions. Dense neural blocks and sparse enzyme dictionary blocks can be stored separately without changing the mathematical score. The method requires the frozen feature extractors, trained encoders, refinement heads, training dictionary, and feature-normalization statistics at deployment. Biological training annotation files are not required at inference.

1.9 Optimization and target-specific fitting

We fit a separate model for each ReactZyme split and for EnzymeMap, using the corresponding target training associations. Trainable retrieval components are freshly initialized for each target; the pretrained feature extractors and SLEEC prior are reused and frozen. In particular, EnzymeMap does not inherit a ReactZyme-trained F3 checkpoint.

Phase 1 uses seed 42, association batch size 512, AdamW learning rate 10^{-4} , weight decay 10^{-2} , and constant learning rate. We use the epoch-10 snapshot. The original ReactZyme trajectories continued to epoch 20; their retained epoch-10 snapshots define this recipe. Features are cached, training uses mixed BF16 precision, and contrastive calculations are kept in FP32. Phase 2 uses seed 42, 100 full-graph AdamW updates, learning rate 10^{-4} , weight decay 10^{-3} , and gradient-norm clipping at 1. The declared step-100 state is used for the reported configuration.

The graph, dictionary centers, and dictionary associations use training endpoints only. The optional biological annotations supply additional supervision on those endpoints and are disclosed separately from the association graph. Matching target training associations therefore does not imply identical total pretrained resources or annotation information across methods. The present recipe was developed through exploratory experiments; the fixed recipe should not be described as a configuration selected without prior inspection of benchmark outcomes.

A Notation and exact layer specifications

Symbol	Meaning
$Y, \mathcal{D}_{\text{tr}}$	Binary target-training association matrix and its unique positive edges.
b_e, b_r	Native 512-dimensional phase-1 protein and reaction outputs.
g_e, f_e	Separately normalized global and fused protein branches.
p_{ki}, a_{ki}	Raw and uniform-smoothed residue attention for learned slot k .
ω_{vj}	Gate weight for protein view v and hidden coordinate j .
s	Learned bounded protein fusion scale; initial value 0.1.
ν	Protein fusion multiplier used after training; value 3.
h_E, h_R	Independent residual heads fitted with frozen F3 inputs.
κ, c	Residual coefficient in training (0.2) and inference multiplier (0.5).
$u_e^{\text{tr}}, u_r^{\text{tr}}$	Refined endpoints used in the phase-2 training objectives.
u_e, u_r	Refined endpoints after the two inference adjustments.
λ_f, ξ_c, w_{ic}	Family coefficient, category weight, and membership confidence; these have different roles.
a_e, a_r	Normalized training-reaction dictionary coordinates.
Φ_E, Φ_R	Complete 512 + N_R endpoint representations.
α	Dictionary contribution to the final dot product; value 0.4.

Table 2: Active neural components in the reported configuration. Dimensions are listed in execution order.

Component	Specification
Frozen residue features	ProtT5; 1024 coordinates per retained residue.
Frozen SLEEC scorer	Linear 1024 $\rightarrow$ 256; ReLU; linear 256 $\rightarrow$ 1.
Residue adapter	LayerNorm(1024); linear 1024 $\rightarrow$ 256; GELU.
Keys and values	Keys: bias-free linear 256 $\rightarrow$ 256 followed by normalization. Values: affine linear 256 $\rightarrow$ 256.
Learned residue queries	Four trainable 256-dimensional vectors; orthogonal initialization; normalized before scoring.
Global/SLEEC view projection	Two independent networks: LayerNorm(1024); linear 1024 $\rightarrow$ 512; GELU; dropout 0.1; linear 512 $\rightarrow$ 256; normalization.
Learned-slot view projection	Four independent networks: LayerNorm(256); linear 256 $\rightarrow$ 512; GELU; dropout 0.1; linear 512 $\rightarrow$ 256; normalization.
Protein gate	Linear 1536 $\rightarrow$ 256; GELU; linear 256 $\rightarrow$ 1536; reshape to 6×256 ; softmax over views.
Global output branch	LayerNorm(1024); linear 1024 $\rightarrow$ 512; GELU; dropout 0.1; linear 512 $\rightarrow$ 512; normalization.
Fused output branch	LayerNorm(256); linear 256 $\rightarrow$ 512; GELU; dropout 0.1; linear 512 $\rightarrow$ 512; normalization.
Uni-Mol2 molecule pooling	Separate scalar affine attention for reactants and products, acting on 768-dimensional molecule vectors.

Component	Specification
ChIRo molecule pooling	Separate scalar affine attention for reactants and products, acting on 256-dimensional molecule vectors.
Reaction modality MLPs	Input widths 768, 3072, 1024, 617 for T5, Uni-Mol2 composition, ChIRo composition, and chemistry. Each follows $d_m \rightarrow 1024 \rightarrow 512 \rightarrow 512$; ReLU after its first two layers.
Reaction token normalization	One LayerNorm(512), shared over modality tokens. No per-token ℓ_2 normalization in this configuration.
Reaction modality attention	Linear $512 \rightarrow 512$; ReLU; linear $512 \rightarrow 1$; masked softmax over available modalities.
Reaction output projection	Linear $512 \rightarrow 1024$; ReLU; linear $1024 \rightarrow 512$; normalization.
Phase-2 heads	Two independent copies of LayerNorm(512); linear $512 \rightarrow 1024$; GELU; linear $1024 \rightarrow 512$. Last affine layer initialized to zero.
Biological neighborhood loss	A nonparametric training helper. Zero additional learned parameters and no inference-time readout.

The two phase-2 heads contain 2,102,272 parameter values in total. Biological supervision changes their gradients but not their parameter names or shapes. The four residue queries are shared across proteins; they are not four trainable vectors created separately for each sequence.

B Attention implementation and sequence preprocessing

Padding is removed from softmax denominators and pooling sums, and its contribution is zero. Before the frozen scorer and residue adapter, padded values are replaced by zero so that padded NaN sentinels do not enter learned linear maps. SLEEC logits are detached. Centering its logits alone would not alter softmax, but centering before clipping gives a protein-relative clipping rule. The use of logits, rather than sigmoid scores, is material to the resulting distribution.

The learned attention scale is initialized by $\rho_\gamma = \text{logit}(4/10)$, and the fusion scalar by $\rho_s = \text{logit}(0.1/0.5)$. The normalization floor in the protein multiview module is 10^{-6} . Attention entropy uses natural logarithms and a probability floor of 10^{-12} inside the logarithm. Regularization in Eqs. (25)–(26) is evaluated on raw learned attention, not on the SLEEC distribution and not on the smoothed distributions used to pool values.

For the F3 protein tower, at most 1022 residue vectors are retained. For a longer cached sequence, the deterministic **ends_center** policy keeps $\lfloor 1022/4 \rfloor$ vectors at each end and the remaining budget from the center, preserving their original order. The dictionary’s enzyme means are exported separately by averaging the stored residue bank; they should not be silently replaced with the neural tower’s potentially truncated mean. EnzymeMap’s specified screening library already restricts sequence length to at most 650 amino acids.

The pooling operations scale linearly in sequence length for a fixed number of slots. In particular, learned residue attention forms an $L \times 4$ matrix, not an $L \times L$ self-attention matrix. This complexity statement concerns the trainable pooling module after frozen features are available; it does not describe the cost of computing ProtT5 features.

C Fixed chemistry descriptor

The 617-dimensional chemistry vector consists of four blocks:

$$617 = 512 + 64 + 4 + 37. \quad (47)$$

The first block averages 512-bit radius-2 Morgan fingerprints of valid participant molecules. The second block applies sum, mean, maximum, and standard-deviation aggregation to 16 molecule descriptors, yielding 64 coordinates. The descriptors are exact molecular weight, heavy-atom count, heteroatom count, formal charge, hydrogen-bond donor count, hydrogen-bond acceptor count, topological polar surface area, logP, rotatable-bond count, ring count, aromatic-heavy-atom fraction, stereocenter count, wildcard-atom count, nitrogen count, oxygen count, and combined phosphorus/sulfur count.

The four set statistics are participant count, unique-participant fraction, valid-participant fraction, and wildcard-containing fraction among valid participants. The final block is a 37-position capacity for the recognized core-cofactor vocabulary fitted on training reactions; unused positions are zero. Descriptor means and standard deviations are fitted on valid training reactions. A standard deviation below 10^{-6} is replaced by one.

Participant canonicalization removes atom-map indices while retaining isomeric SMILES information. When both sides of a stored self-reaction have the same canonical participant multiset, the participant list is used once. Otherwise, the two sides are combined for this descriptor. Thus this chemistry block is a participant-set feature, even when the neural molecular branches retain physical reaction sides. Existing cofactor-related input coordinates are distinct from the optional cofactor *loss* in Section 1.6; the baseline does not exclude all biological information.

D Efficient evaluation of the biological objective

For one category, define the weighted sums

$$A_c = \sum_{i \in I_c} w_{ic}, \quad Q_c = \sum_{i \in I_c} w_{ic}^2, \quad s_c = \sum_{i \in I_c} w_{ic} \hat{z}_i, \quad (48)$$

and the diagonal correction $V_c = \sum_{i \in I_c} w_{ic}^2 \|\hat{z}_i\|_2^2$. The ordered off-diagonal confidence mass is $T_c = A_c^2 - Q_c$. Expanding the dot product of the weighted sum gives

$$\sum_{\substack{i,j \in I_c \\ i \neq j}} w_{ic} w_{jc} (1 - \hat{z}_i^\top \hat{z}_j) = T_c - \|s_c\|_2^2 + V_c. \quad (49)$$

Retaining V_c makes the expression exact even when numerical normalization does not return exactly unit norm. For the background, define

$$A_c^{\text{out}} = \sum_{j \in B_c} \bar{w}_j^f, \quad s_c^{\text{out}} = \sum_{j \in B_c} \bar{w}_j^f \hat{z}_j. \quad (50)$$

Then $U_c = A_c A_c^{\text{out}}$ and

$$\sum_{i \in I_c} \sum_{j \in B_c} w_{ic} \bar{w}_j^f (1 - \hat{z}_i^\top \hat{z}_j) = U_c - s_c^\top s_c^{\text{out}}. \quad (51)$$

Background sums are obtained from the total family-annotated sum after subtracting the category members’ contributions using their *family* maximum confidence. This is important when a member’s confidence for category c differs from its largest confidence in the family.

The implementation uses indexed reductions to evaluate these identities. If M_f is the number of annotated memberships in a family and C_f its number of categories, their additional time is proportional to $(N_X + M_f + C_f)d$, for representation width $d = 512$; it does not require a dense $N_X \times N_X$ biological distance matrix. This saving does not remove the separate $N_R \times N_E$ matrix used by the full-graph association loss.

Distinction from the earlier attraction-only objective. An earlier ablation used

$$\ell_c^{\text{attraction}} = \frac{T_c - \|s_c\|_2^2 + V_c}{n_c(n_c - 1)}. \quad (52)$$

This encourages absolute within-category compactness. The reported `all_1` recipe instead uses the relative hinge in Eq. (36). These objectives should not be interchanged when describing or reproducing the biological experiments.

E Why the two association losses differ

For a single phase-1 positive, write $D_{ap} = \exp(z_{ap}) + \sum_{n \in N_B(a)} \exp(z_{an})$. Its gradients are

$$\frac{\partial \ell_{\text{dec}}(a, p)}{\partial z_{ap}} = \frac{\exp(z_{ap})}{D_{ap}} - 1, \quad (53)$$

$$\frac{\partial \ell_{\text{dec}}(a, p)}{\partial z_{an}} = \frac{\exp(z_{an})}{D_{ap}}, \quad n \in N_B(a). \quad (54)$$

Other known positives are absent from this positive’s denominator and have zero derivative in this individual term. They receive their own positive terms. Averaging these terms within an anchor ensures that adding more recorded positives does not simply multiply its loss weight.

Phase 2 instead fits a categorical distribution with target probability $1/d_a$ for each of an anchor’s d_a positives. Its per-logit derivative is the predicted probability minus the target probability. Its minimum cross-entropy for unconstrained logits approaches $\log d_a$, the entropy of the uniform-positive target, rather than zero when $d_a > 1$. Consequently, numerical loss values from the two phases are not directly comparable. Both losses use unobserved associations as contrastive alternatives; neither resolves all missing-positive annotation.

F Annotation provenance and benchmark-specific inputs

The biological categories are built on the training endpoints of each target. For EnzymeMap, EC assignments and atom mapping are read from the native EnzymeMap cache only at indices recorded in the official training associations. Mechanism labels are derived from the native mapped reactants and products. The native quality field must be at least 0.5, and the retained confidence is capped at one. Cofactor presence is recognized from the training reaction participants and assigned confidence 0.4.

For ReactZyme, native EC annotations are restricted to training protein sequences. Training reaction EC annotations use reaction-only Rhea matching and exact canonical-chemistry lookup. Coarse bond-change categories use the cached mapping of each training reaction, retain mapping confidence at least 0.5, and multiply this confidence by the reaction-matching confidence. Cofactor categories use the permitted training participant sets. These reaction-only annotation resources

provide information beyond the raw released participant-set input; they are auxiliary training supervision, not an inference-time reconstruction of missing reaction direction.

For every training edge, the reaction’s observed categories are added to its associated protein, retaining the maximum confidence when a membership is supported more than once. No held-out reaction–enzyme association is used for this propagation. Unknown or absent categories are omitted. In the current EnzymeMap annotation vocabulary, the observed coarse mechanism groups are phosphate transfer, redox/carbonyl interconversion, stereochemical rearrangement, and sulfur/thiol chemistry. These broad labels should not be described as a detailed model of catalytic mechanisms.

The target training graph and dictionary sizes are given in Table 3. Duplicate association rows are removed when building the compact phase-2 graph. Training endpoints can remain eligible candidates under a benchmark’s candidate-pool definition; the method does not silently remove them.

Target	Training rows	Unique phase-2 edges	Reaction anchors
ReactZyme Reaction-Sim	147,393	147,393	6,977
ReactZyme Enzyme-Sim	152,748	152,748	7,386
ReactZyme Time	149,554	149,554	7,281
EnzymeMap	34,427	34,180	12,603

Table 3: Target-specific training inputs for the implemented recipe. Each row has independently fitted retrieval weights, phase-2 heads, and a dictionary.

G Optimization recipe and executable order

Setting	Value
Phase-1 seed; association batch size	42; 512
Retained phase-1 snapshot	End of epoch 10
Phase-1 optimizer	AdamW, lr = 10^{-4} , weight decay 10^{-2}
Phase-1 inverse temperature	Fixed $\beta = 5$
Phase-1 directional weights	1/2 for each direction
Unknown-negative weight	1
Entropy and diversity coefficients	0.01 and 0.01
Protein attention uniform mixture	$\eta = 0.05$
Gate uniform mixture	$\zeta = 0.05$
Phase-2 optimizer	AdamW, lr = 10^{-4} , weight decay 10^{-3}
Phase-2 updates; seed	100 full-graph updates; 42
Phase-2 temperature; identity coefficient	$\tau_2 = 0.2$; 10
Phase-2 residual training coefficient	$\kappa = 0.2$
Phase-2 gradient clipping	Global gradient norm 1
Biological mode; margin	Relative; $m = 0.1$
Biological family coefficients, <code>all_1</code>	$(\lambda_{\text{EC}}, \lambda_{\text{cof}}, \lambda_{\text{mech}}) = (1, 1, 1)$
Biological family divisor	Fixed at 3, including removal controls
Phase-1 biological supervision, <code>all_1</code>	None
Inference protein fusion multiplier	$\nu = 3$
Inference residual multiplier	$c = 0.5$

Setting	Value
Dictionary score coefficient	$\alpha = 0.4$
Protein dictionary neighbors	32
Dictionary kernel temperatures	$\tau_E = \tau_R = 0.03$
Reaction dictionary truncation	None; all training reaction anchors

Training and inference procedure.

1. Construct the target training association graph and the frozen feature caches. Initialize the trainable retrieval components for that target and load the frozen pretrained feature resources and SLEEC scorer.
2. Train the phase-1 encoders with Eq. (27). For each batch, deduplicate endpoint IDs and recover all known in-batch training positives. Retain the epoch-10 state.
3. Freeze F3. Export native training embeddings using its learned scale s , without the factor-three inference adjustment. Build the dictionary centers, training protein bank, training reaction bank, and association map.
4. Initialize the residual heads at identity. For each of 100 updates, encode every training endpoint, evaluate the full bidirectional cross-entropy and identity loss, and optionally add the biological relative loss. Clip gradients and update only the residual heads.
5. At inference, obtain the protein’s global and fused branches and use νs in Eq. (43). Apply the trained heads with residual coefficient $c\kappa = 0.1$.
6. Independently compute the enzyme and reaction dictionary coordinates from frozen raw features and the stored training dictionary. Concatenate the weighted blocks, or retain equivalent dense and sparse blocks.
7. Rank candidates by Eq. (46) in the requested direction. No candidate labels, candidate-pool normalization, or biological annotation files are needed for encoding new endpoints.

Checkpoint representation of the fusion adjustment. The EnzymeMap deployment checkpoint already stores the factor-three change to the bounded fusion scalar, so its runtime fusion multiplier is one. The ReactZyme deployment checkpoints store the native scalar and apply three at runtime. These represent the same recipe. Applying three again to the calibrated EnzymeMap checkpoint would implement a different model. Phase-2 training used the native feature exports for every target.

Numerical evaluation. Dictionary feature norms, centers, and nearest-neighbor similarities use FP64 reductions where implemented before storing FP32 features. Neighbor selection resolves cutoff ties by the lower training-anchor index. Equivalent mathematical score decompositions can nevertheless have different floating-point rounding when executed in different orders. The frozen evaluation routines and candidate ordering should therefore be retained, particularly for nearly tied ReactZyme reaction candidates. The dictionary is part of the deployed method, not a post-hoc normalization fitted to the current screening library.

H Scope of the biological extension

The main specification above is the phase-2-only biological configuration. Experiments that retrain F3 with biological supervision are a different variant: they add the family-averaged loss to the phase-1 objective, using deduplicated minibatch endpoints rather than the full graph. In that case the gradient can reach the learned residue queries and reaction encoder. Category eligibility and background coverage then depend on each minibatch, so this is not equivalent to adding the same nominal coefficient only in phase 2.

The architecture and loss definition describe how biological relationships are introduced, not whether they improve generalization. That empirical question requires matched controls, including removal and shuffled-label experiments. Equal family coefficients do not imply equal gradient influence or equal predictive benefit. The 512 neural coordinates have no fixed biochemical names; only the dictionary coordinates have an explicit training-reaction interpretation.

References

- [1] Keir Adams, Lagnajit Pattanaik, and Connor W. Coley. Learning 3d representations of molecular chirality with invariance to bond rotations. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=hm2tNDdgaFK>.
- [2] Ahmed Elnaggar, Michael Heinzinger, Christian Dallago, Ghalia Rehawi, Yu Wang, Llion Jones, Tom Gibbs, Tamas Feher, Christoph Angerer, Martin Steinegger, Debsindhu Bhowmik, and Burkhard Rost. ProtTrans: Toward understanding the language of life through self-supervised learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 44(10):7112–7127, 2022. doi: 10.1109/TPAMI.2021.3095381. URL <https://arxiv.org/abs/2007.06225>.
- [3] Chengyue Gong, Daiheng Zhang, Jeffrey Ouyang-Zhang, Qiang Liu, Adam Klivans, and Daniel J. Diaz. Enzyme classification via semi-supervised functional residue learning. *bioRxiv*, 2026. doi: 10.64898/2026.02.11.705200. URL <https://doi.org/10.64898/2026.02.11.705200>.
- [4] Xiaohong Ji, Zhen Wang, Zhifeng Gao, Hang Zheng, Linfeng Zhang, Guolin Ke, and Weinan E. Uni-Mol2: Exploring molecular pretraining model at scale. In *Advances in Neural Information Processing Systems*, volume 37, 2024. URL <https://arxiv.org/abs/2406.14969>.
- [5] Tatsuya Sagawa. ReactionT5v2: Official model repository. <https://github.com/sagawatatsuya/ReactionT5v2>, n.d. Accessed September 21, 2026.
- [6] Tatsuya Sagawa and Ryosuke Kojima. ReactionT5: A large-scale pre-trained model towards application of limited reaction data. *arXiv preprint arXiv:2311.06708*, 2023. URL <https://arxiv.org/abs/2311.06708>.